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Ultra precision positioning system for servo motor–piezo actuator using the dual servo loop and digital filter implementation

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Abstract

The ultra precision positioning technique has become one of the important parts in the development of precision machines. For positioning systems having a long stroke with ultra precision, a combined system including a global stage (coarse stage) and a micro stage (fine stage) is designed in this paper. A ball screw based servo motor is used as the global stage and a piezo actuator as the micro stage, in which some design parameters are analysed. To improve positional accuracy and remove the noise components of motion, the digital Chebyshev filter has been designed and implemented. The implemented digital filter has been found to improve accuracy by as much as three times. Positional accuracy has been readily achieved within 10 nm over the typical 200 mm stroke, and therefore this technique can be applied to develop high precision semiconductor equipment such as lithography steppers and probers. © 2000 Elsevier Science Ltd. All rights reserved.

Keywords: Ultra precision; Global coarse stage; Micro fine stage; Dual servo; Digital filter

1. Introduction

In recent years, the ultra precision positioning technique has become an important part of the development of precision machines. The issue of achieving sub-micrometer accuracy over a range of a few hundred millimeters has been mainly targeted in the field of ultra precision machine tools and semiconductor manufacturing equipment such as lithographers and steppers. One of the most practical methods for achieving this is to use the coarse and the fine stages for precision

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positioning [1–3]: the coarse stage (global stage) covers the long range up to an accuracy of a few micrometers, and the fine stage (micro stage) covers the small range of sub-micrometer accuracy, in which an appropriate control algorithm is desired for linking the stages. Sakuta et al. [1] developed a dual servo mechanism which was composed of a friction drive and a PZT actuator. The PZT actuator was mounted on the friction drive slide, and the positioning system was controlled using a kind of inchworm movement method. Okazaki et al. [2] developed a positioning stage using a fine/coarse servo system. The displacement of the stage was measured with a laser interferometer and gap gage, where the servo system was controlled with task sharing in the displacement range and in the frequency range. Lee and Kim [3] developed a dual servo stage mechanism, which was composed of a 3-DOF linear motor driven global stage and a 3-DOF PZT actuated fine motion stage, where the mechanism was controlled with the PID scheme. In this paper, an ultra precision positioning system has been developed: the combination of a global stage (coarse stage) and a micro stage (fine stage) for a positioning system having a long stroke with ultra precision. A ball screw based servo motor is used for the global stage, and the PZT actuator driven micro stage is designed using the design principle of the flexure hinges, in which some design parameters are readily evaluated. For the high static/dynamic performance, the dual servo control loop algorithm has been implemented to link the global and micro stages, which gives sub-micrometer accuracy over the long range of motion. The environmental effects such as vibrations/noises have been found to be the main obstacles for achieving higher accuracy in the sub-micrometer range, and some specific digital filtering techniques have been implemented. The Chebyshev digital filtering technique has been designed and implemented after diagnosing the vibration sources. The developed system has been successfully applied to a practical positioning system, giving at least three times the level of accuracy over the long range travel of about 200 mm, demonstrating the high performance with ultra precision.

2. Design of micro stage for fine motion

An efficient method for achieving fine motion with ultra precision is to use the flexure hinge mechanism. The flexure hinge mechanism uses the elastic hinge, thus providing smooth and continuous motion without any friction. Fig. 1 shows two typical flexure hinges: a notched hinge

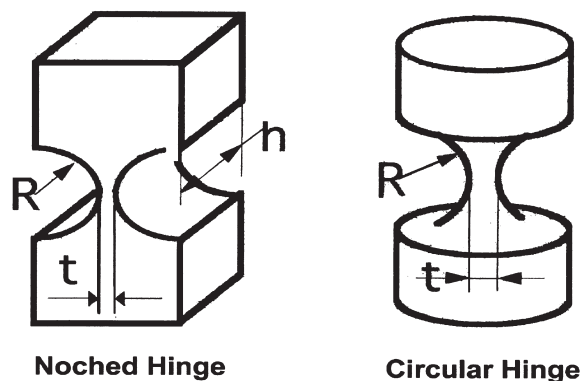


Fig. 1. Flexure hinges: notched flexure, circular flexure.

and a circular flexure, where the circular flexure is known as the universal joint due to its universality in the motion performance.

In the notched hinge of Fig. 1, where R is the notch radius, t is the thickness, and w is the width of the notched hinge, the bending stiffness K_B of the notched hinge is formulated as per Eq. (1), where $2R+t$ almost equals the notch width, w . The bending stiffness of the circular flexure is similarly formulated as given in Eq. (2), where t and R are similarly defined as in Fig. 1, and E is the Young's modulus of the flexure material [4,5].

$$K_B = \frac{2Eht^{5/2}}{9\pi R^{1/2}} \text{ (notched hinge)} \quad (1)$$

$$K_B = \frac{Et^{7/2}}{20R^{1/2}} \text{ (circular hinge)} \quad (2)$$

For the design of the micro stage using the hinge flexures, multiple hinges are generally used in order to give a symmetrical structure and little parasitic motion. Thus, in this paper the micro fine stage is designed using eight notched hinges for providing one degree of freedom of motion. Fig. 2 shows the designed micro stage, and the static stiffness of the micro stage is evaluated as follows. Let q represent the displacement along the direction of motion of the stage, and L the effective length between the center of the stage and the center of respective circular hinges. Then each hinge rotates by the amount of $q/2L$, and thus the total elastic energy of the micro stage becomes the sum of the elastic energy of the hinges. When n is the total number of the hinges, the total elastic energy, U , becomes,

$$U = \sum_{i=1}^n \frac{1}{2} K_B \frac{q^2}{(2L)^2} = \frac{n}{2} K_B \frac{q^2}{(2L)^2} = K_B \frac{q^2}{L^2} \quad (3)$$

where the number of hinges is eight.

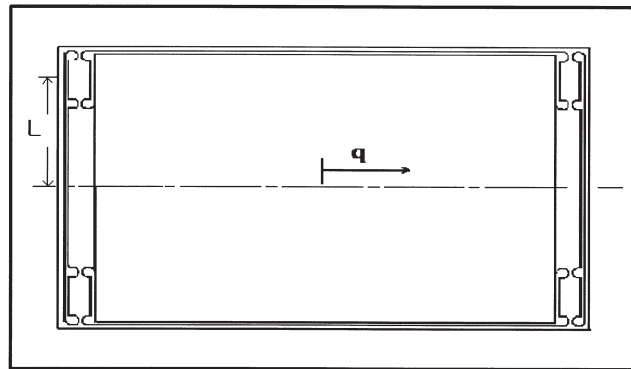


Fig. 2. Designed micro stage.

Therefore, the elastic force F_q along the q direction is derived from the partial derivative of the total energy, U , with respect to q , from the energy principle. That is,

$$F_q = \frac{\delta U}{\delta q} = 2K_B \frac{q}{L^2} \quad (4)$$

Therefore, the static stiffness, λ_q of the micro stage along the q direction (direction of motion) is evaluated as

$$\lambda_q = \frac{F_q}{q} = \frac{2K_B}{L^2} = \frac{4Ebt^{5/2}}{9\pi L^2 R^{1/2}} \quad (5)$$

The dynamic characteristics can also be evaluated by applying the Lagrangian principle [6]. When q' , q'' are the velocity and acceleration of the stage along the q direction, the total kinetic energy of the stage, T , is

$$T = \frac{1}{2}Mq'^2 + 4\frac{1}{2}I_0\left(\frac{q'}{L}\right)^2 \quad (6)$$

where M is the mass of the stage and I_0 is the moment of inertia of the body of the flexure hinge with respect to the center of the flexure body. Thus

$$I_0 = \frac{ml^2}{12} + m\left(\frac{l}{2}\right)^2 = \frac{ml^2}{3} \quad (7)$$

where m is the mass of the body of the flexure hinge and l is the length of the body of the flexure. Therefore, from Eq. (6),

$$T = \frac{1}{2}Mq'^2 + 4\frac{1}{2}\frac{ml^2}{3}\left(\frac{q'}{L}\right)^2 = \frac{1}{2}Mq'^2 + \frac{2ml^2q'^2}{3L^2} \quad (7a)$$

On the other hand, the elastic potential energy, U , can be derived from the stiffness, λ_q , and the displacement, q , along the direction of motion. That is,

$$U = \frac{1}{2}\lambda_q q^2 \quad (8)$$

Therefore, the Lagrangian multiplier, L , can be defined as

$$L = T - U = \frac{1}{2}Mq'^2 + \frac{2ml^2q'^2}{3L^2} - \frac{1}{2}\lambda_q q^2 \quad (9)$$

and the equation of motion of the stage becomes,

$$\frac{d}{dt}\left(\frac{\delta L}{\delta q'}\right) - \frac{\delta L}{\delta q} = 0 \quad (10)$$

In Eq. (10), terms are derived from Eq. (9), that is,

$$\frac{\delta L}{\delta q'} = Mq' + \frac{4ml^2 q'^2}{3L^2}, \quad \frac{\delta L}{\delta q} = -\lambda_q q$$

Thus, from Eq. (10)

$$\left(M + \frac{4ml^2}{3L^2}\right)q'' + \lambda_q q = 0 \quad (10a)$$

Therefore, the natural frequency, ω_n , of the stage is derived such as

$$\omega_n = \sqrt{\frac{\lambda_q}{M + \frac{4ml^2}{3L^2}}} = \sqrt{\frac{4Ebt^{2/5}}{9\pi R^{1/2}L^2 \left(M + \frac{4ml^2}{3L^2}\right)}} = \sqrt{\frac{4Ebt^{2/5}}{9\pi R^{1/2}L^2 M + 12\pi R^{1/2}ml^2}} \dots (\text{rad/s}) \quad (11)$$

Therefore, the static and dynamic performance of the designed stage is assessed for the micro stage of fine motion.

3. Dual servo control system

In order to provide the long range motion, the designed micro stage is linked with the global stage, where the global stage is designed with the ball screw based servo motor. Both the global stage and the micro stage are combined mechanically and an appropriate control algorithm is implemented for the smooth linking between the two motions of the global–micro stages. In this paper, the dual servo loop control algorithm is implemented for that purpose and a detailed block diagram of the control scheme is shown in Fig. 3. The global stage unit consists of the ball screw

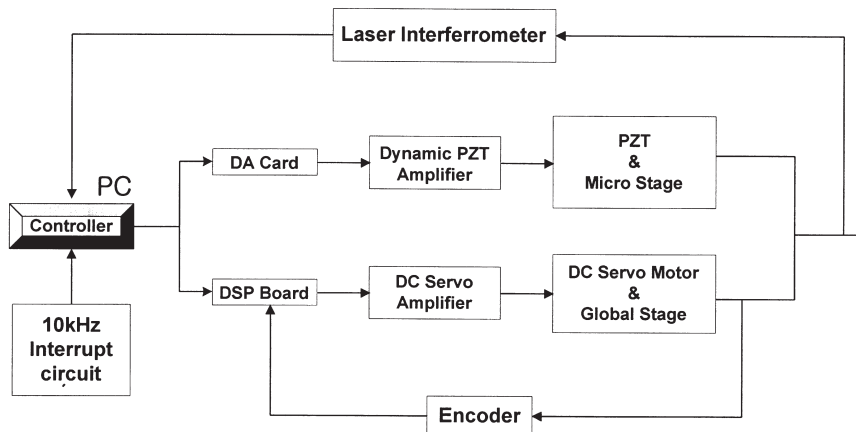


Fig. 3. Block diagram of the dual servo loop for the positioning system.

based DC servo motor, a DC servo amplifier, a rotary encoder for feedback, and the DSP (digital signal processing) board for the PID control of the coarse motion. The micro stage unit consists of the PZT actuator, a dynamic PZT amplifier, a laser interferometer for the position feedback, and the D/A (digital to analogue) interface card for the motion control of the fine motion. The PC functions as the host controller for the global–micro stage system, in which the 10 kHz based timer/counter interrupt circuit and associated interrupt routines are also implemented. This is in order to give the precision sampling time for the precision motion control. The dual servo loop algorithm is successfully implemented and detailed procedures are as given in Fig. 4. When the system target position is issued from the controller (PC), the DSP board of the global stage takes the command, giving pulses for movement of the servo motor engaged in the global stage. The movement of the global stage is then measured by the installed laser interferometer at a 10-kHz rate, making a closed feedback control loop for the global stage. When the measured movement

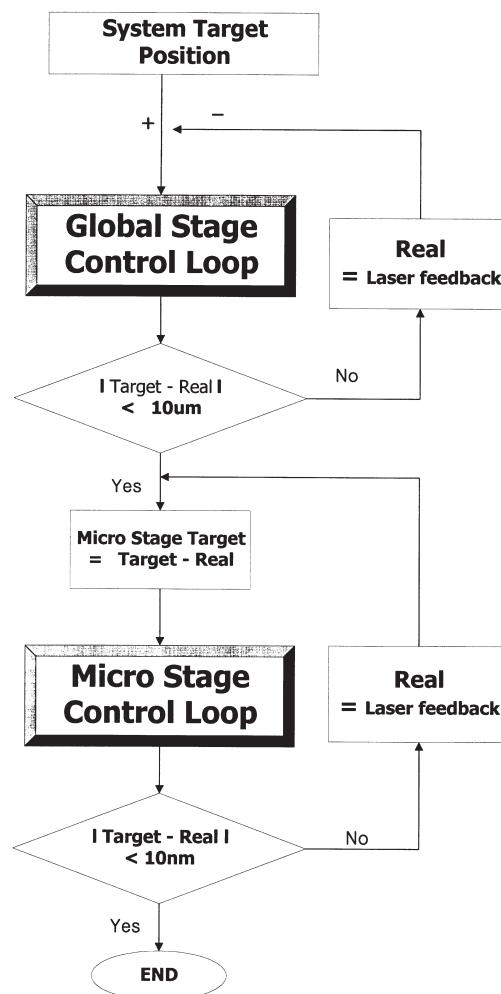


Fig. 4. Flow chart for the dual servo loop algorithm.

of the stage is close to the target position, about 10 μm , which is the average positional accuracy of the global stage, the controller of the micro stage takes the motion control. Thus, the PC gives commands to the D/A interface card, generating the analogue signal to the PZT amplifier followed by the micro stage actuation. The measured data from the laser interferometer also provide the feedback control loop for the micro stage movement.

4. Digital filter implementation for improvement of positional accuracy of the global/micro stages

Though the dual servo control system has been implemented for the motion control, there have been several factors degrading the positional accuracy of the stage system, such as vibration and noise. Fig. 11(a) shows the stage movement under 500 nm command, in which a noise level of about 30 nm has been found, although the laser interferometer measurement system was upgraded to 5 nm resolution. For the vibration/noise source diagnosis, the response of the stage system is measured with suitable vibration pickups, and the frequency spectrum of the vibration/noise components are analyzed as shown in Fig. 5, where the components at about 46 Hz are dominant over the full frequency spectrum range. The dominant noise components are diagnosed as the vibration of the optics holder fastened to the magnetic chuck on the table (shown in Fig. 6), even though the magnetic chuck based holder was tightly fastened to the vibration isolation table. Even very small vibrations of the holder seem to be magnified to give that noise level of laser measurement via the relatively long offset of the laser measurement line from the air table. As the noise level of the stage positioning is found to be mainly due to the vibration of the chuck based holder, the accuracy of the stage positioning can be highly improved if appropriate noise filtering techniques are employed. In this paper, the Chebyshev digital filtering technique [7] is applied to remove the 46 Hz noise component, as the digital filtering techniques are widely used for the noise filtering processes. Fig. 7 shows the block diagram of the Chebyshev filter, where $X(z)$ and $Y(z)$ are the input and output signals, respectively, and $H(z)$ indicates the Chebyshev digital filter. The general form of the Chebyshev filter of the n th order is given as follows,

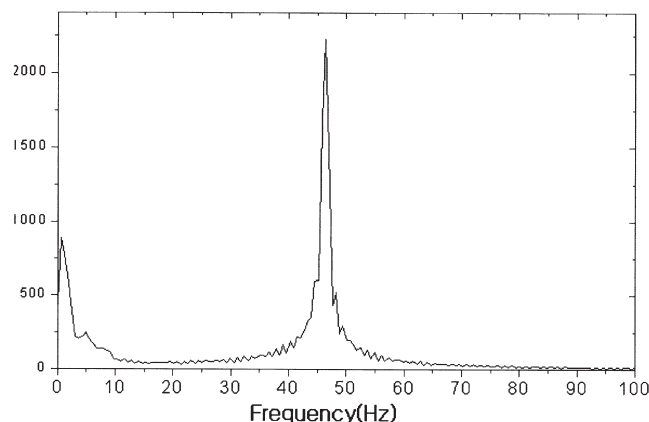


Fig. 5. Frequency spectrum of the stage response.

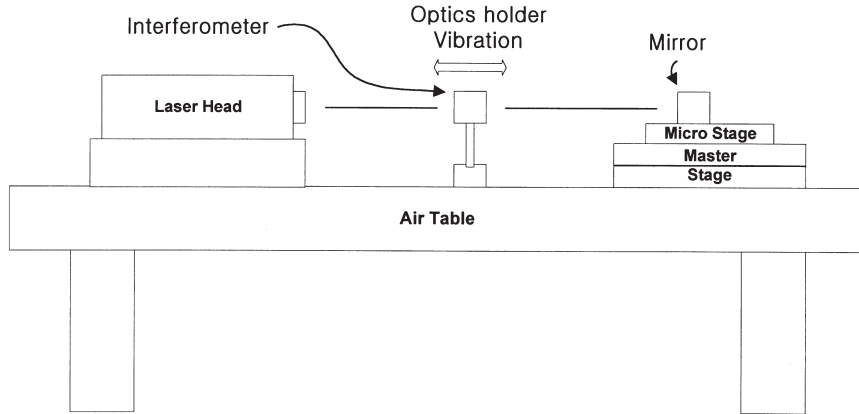


Fig. 6. Measurement system setup demonstrating the optics holder vibration.

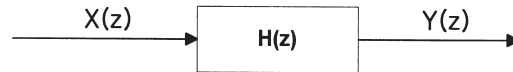


Fig. 7. Block diagram of the Chebyshev filter.

$$H(z) = K \frac{1 + b_1 z^{-1} + \dots + b_n z^{-n}}{1 + a_1 z^{-1} + \dots + a_n z^{-n}}, \quad K = b_0 \quad (12)$$

where K , a_i and b_i are the coefficients to be determined for the filter implementation. In this research, the 10th order Chebyshev filter is considered, which can be decomposed into the products of the five consecutive digital filters of the second order. That is,

$$H(z) = KH_1(z)H_2(z)H_3(z)H_4(z)H_5(z) \quad (13)$$

where $H_1(z)$ to $H_5(z)$ indicate the second order digital filters, and the block diagrams of the respective second order digital filters are shown in Fig. 8, in which all the second order filters and the relevant coefficients are illustrated. The digital filter of Fig. 8 can be represented as the discrete forms for the numerical computation as follows. Let $X_i(k)$ ($i=1, 2, 3, 4, 5$) be the signal at time k , and $X_i(k-n)$ be the signal at the time $(k-n)$, which is n sampling time period before the time k . Remembering that the Z transformation of a function $f(k-n)$ equals $Z^{-n}F(Z)$, where $F(Z)$ is the Z transformation of function $f(k)$, the input/output relationships and the decomposed second order filters can be represented as

$$\begin{aligned} X_1(k) &= -a_{11}X_1(k-1) - a_{12}X_1(k-2) + X(k) + b_{11}X(k-1) + b_{12}X(k-2) \\ X_2(k) &= -a_{21}X_2(k-1) - a_{22}X_2(k-2) + X_1(k) + b_{21}X_1(k-1) + b_{22}X_1(k-2) \\ X_3(k) &= -a_{31}X_3(k-1) - a_{32}X_3(k-2) + X_2(k) + b_{31}X_2(k-1) + b_{32}X_2(k-2) \\ X_4(k) &= -a_{41}X_4(k-1) - a_{42}X_4(k-2) + X_3(k) + b_{41}X_3(k-1) + b_{42}X_3(k-2) \\ X_5(k) &= -a_{51}X_5(k-1) - a_{52}X_5(k-2) + X_4(k) + b_{51}X_4(k-1) + b_{52}X_4(k-2) \\ Y(k) &= KX_5(k) \end{aligned} \quad (14)$$

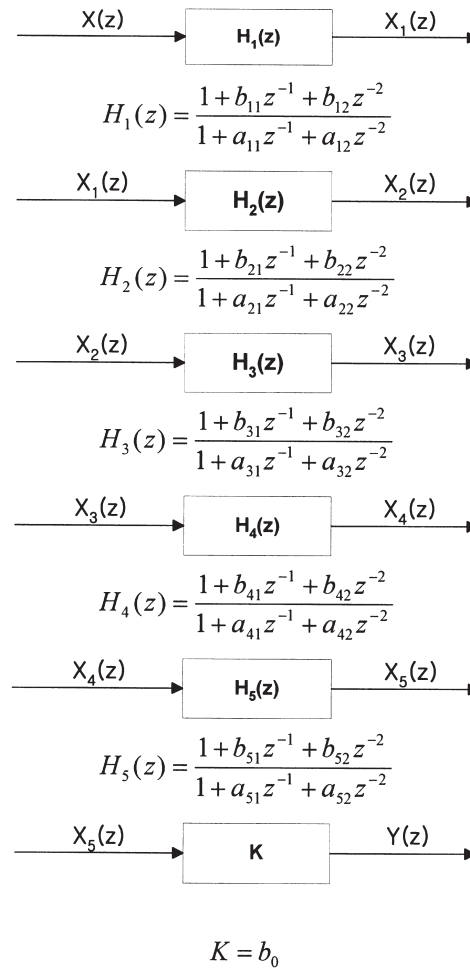


Fig. 8. Decomposed digital filters of the second order, $H_1(z)$ to $H_5(z)$.

where $X_1, X_2, X_3, X_4, X_5, Y$ are the input and output signals to the respective filters as shown in Fig. 8. The five consecutive second order digital filters are evaluated by applying the MATLAB software [8, pp. 2–10], where the cutoff frequency range is selected between 40 and 55 Hz, and the ripple is allowed to be 0.3. The coefficients of the digital filters are successfully calculated up to four digits of floating points, as given in Table 1. The characteristics curve of the calculated Chebyshev digital filter is illustrated in Fig. 9, demonstrating a very narrow notch filter pattern around the frequency range of 46 Hz. The calculated Chebyshev digital filter is implemented to the stage control system, and the block diagram of the modified dual servo loop is shown in Fig. 10, where the Chebyshev filter, $H(s)$, is added on the loop. The PID (proportional integrative derivative) control algorithm is implemented for the global stage motion, and the PI (proportional integrative) control algorithm for the micro stage motion. The gains and constants for the control loop have been determined by some experimental procedures. Fig. 11(a,b) shows the comparison of the step responses with and without the Chebyshev digital filter implementation, when the 500

Table 1
Digital Chebyshev filter coefficients

i	a_{ij}		b_{ij}		K
	$j=1$	$j=2$	$j=1$	$j=2$	
1	-1.6658	0.8483	-1.8901	1.000	0.9782
2	-1.7900	0.9681	-1.8901	1.000	
3	-1.6080	0.7015	-1.8901	1.000	
4	-1.8537	0.9116	-1.8901	1.000	
5	-1.9147	0.9802	-1.8901	1.000	

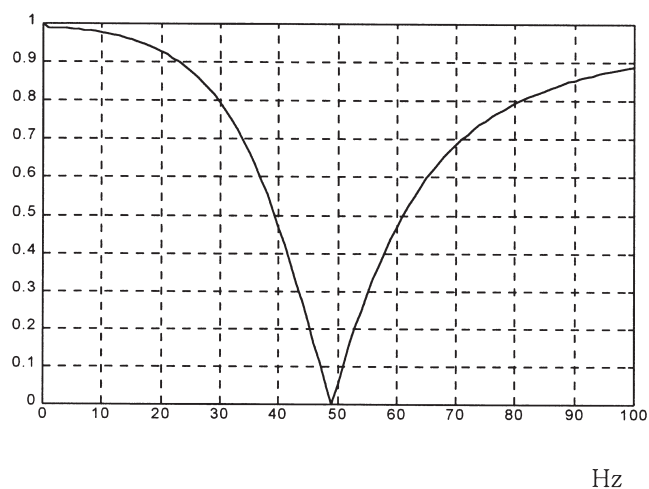


Fig. 9. Characteristic curve of the implemented digital Chebyshev filter.

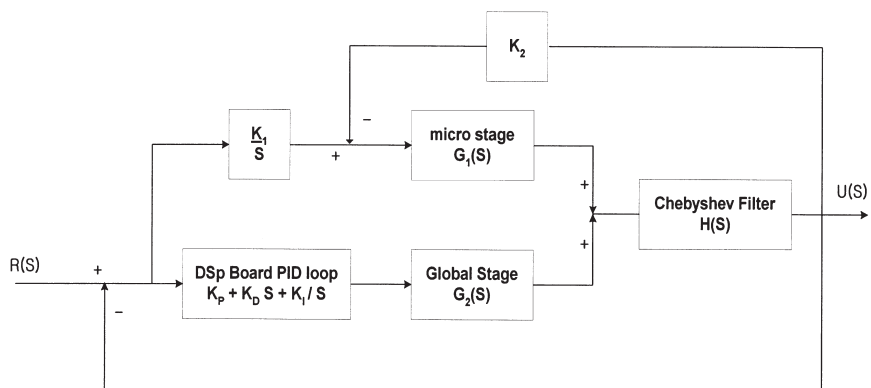


Fig. 10. Block diagram for the dual servo loop implementing the digital filter.

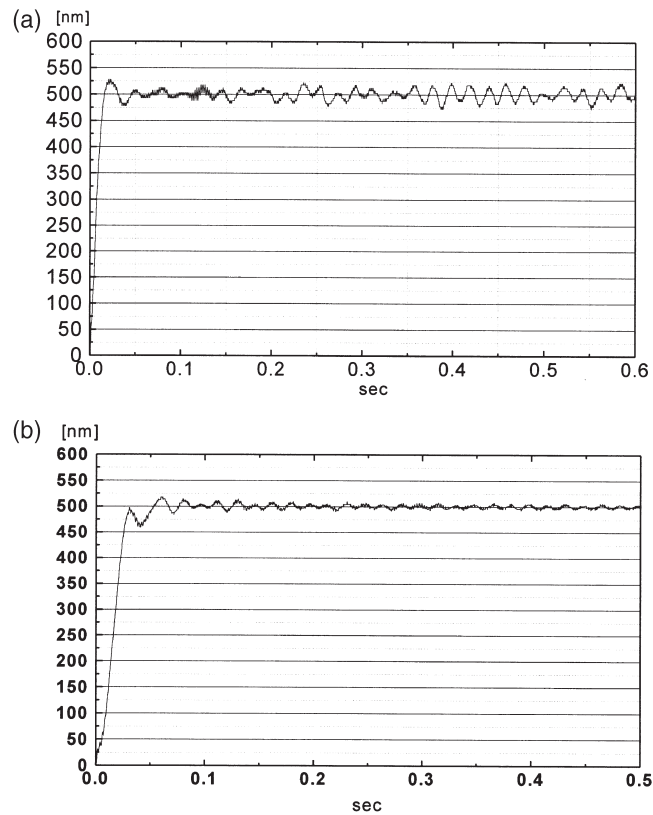


Fig. 11. Step response of 500 nm (a) without the Chebyshev filter and (b) with the Chebyshev filter.

nm step command is given to the stage system. The precision positioning capability is improved from a dispersion level of about 30 nm to about 10 nm, which indicates an improvement in the positioning accuracy of about three times. In order to assess the dual servo performance, a 100 μm command is given to the stage system, and the stage responses are compared for the cases with and without the micro stage actuation. Fig. 12(a) shows the stage response under the 100 μm command when only the global stage is actuated, where a systematic error of about 3 μm is recorded with some fluctuation in the noisy components. Fig. 12(b) shows the stage response when both stages (global and micro stages) are actuated under the 100 μm command, where the micro stage is successfully compensating for the systematic error with very few noisy components. Therefore, the performance of the stage implementing the dual servo stage system is successfully demonstrated, and the positioning accuracy is greatly improved with the introduction of the dual servo control system. The performance of the Chebyshev digital filter is also demonstrated, where the dispersion of the stage motion is greatly reduced from the vibration or noisy environment around the stage setup.

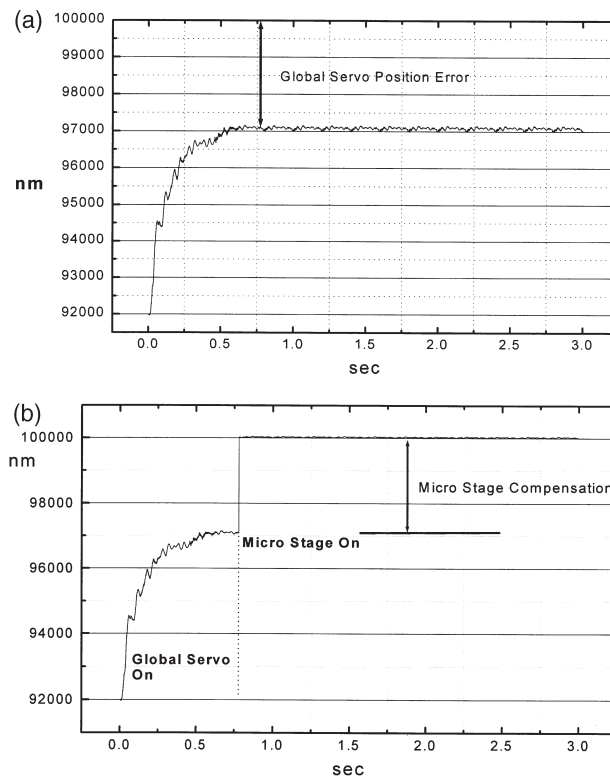


Fig. 12. (a) Response of the global stage under 100 μm command showing 3 μm positional error with noise components. (b) Response of the global-micro stage under 100 μm command showing about 10 nm noise components.

5. Conclusion

An ultra precision positioning system has been developed, comprising the lead screw based servo motor and the flexure based PZT actuator. The conclusions are as follows.

1. An ultra precision positioning system has been developed, comprising a lead screw based servo motor and a flexure based PZT actuator. The dual servo control algorithm has been implemented for the stage control system, and thus about 10 nm positioning accuracy has been achieved over the typical 200 mm stroke.
2. A micro stage has been designed for fine motion, comprising the flexure based PZT actuator. Some relevant design parameters are also assessed and evaluated, including the static and dynamic performance of the stage.
3. In order to improve the positioning accuracy of the stage system, the frequency spectrum of the stage response is analyzed to identify the major vibration sources. As a result, the magnetic chuck based optics holder was identified as the main source of vibration, which is magnified through the offset between the laser measurement line and the table.
4. The Chebyshev digital filter is designed for reducing the vibration of the stage, of which parameters and coefficients of the 10th order are optimized and evaluated with the MATLAB pack-

age. Thus, an improvement in the positioning accuracy of about three times has been achieved, from about 30 nm dispersion (before) to 10 nm dispersion (after).

5. The performance of the dual servo control system has been assessed for the stage system with and without the micro stage. As a result, a significant systematic error (about 3 μm) with some dispersion (about 30 μm) has been observed for the case without the micro stage under the specific 100 μm command, while a virtually negligible systematic error was achieved with about 10 nm dispersion.
6. The performance and efficiency of the developed system, therefore, has been demonstrated, and this technique can be applied to the various ultra precision engineering fields such as semiconductor manufacturing equipment or precision machine tools.

Acknowledgements

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